

$$\Rightarrow \boxed{\nabla \cdot \vec{D} = \rho(r)}$$

Gauss' law
Macroscopic

Deduce Poisson equation

$$\vec{D} = \epsilon \vec{E} \quad E = -\nabla U$$

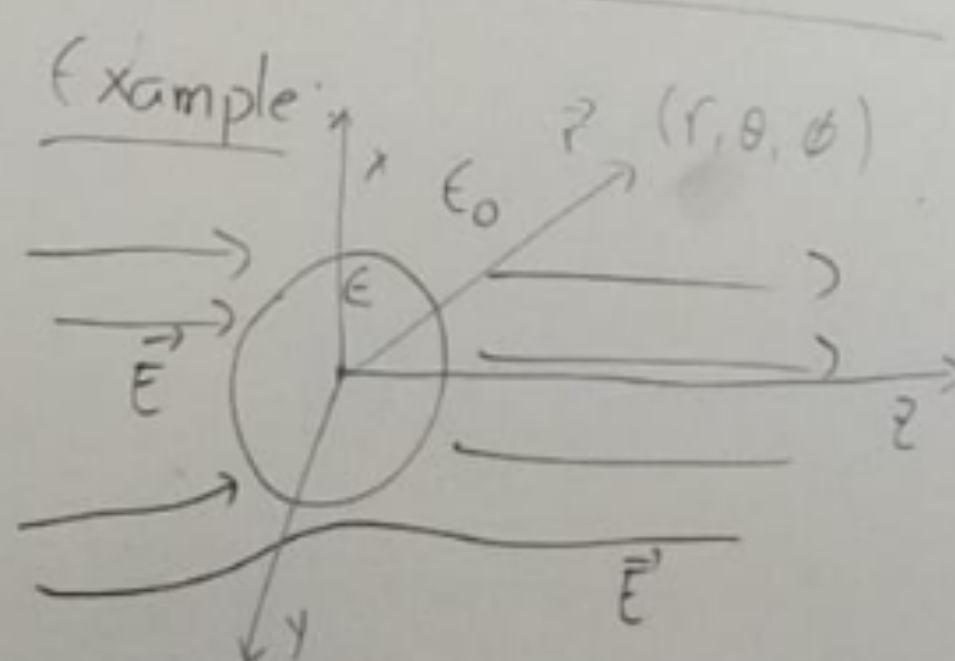
$$\nabla \cdot \vec{D} = \nabla \cdot \epsilon \vec{E} = \epsilon \nabla \cdot \vec{E} = -\epsilon \nabla \cdot \nabla U = \rho$$

$$\Rightarrow \boxed{\nabla^2 U = -\frac{\rho}{\epsilon}}$$

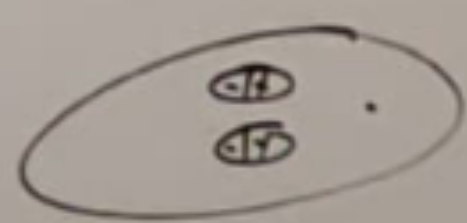
Poisson equation
for dielectric media

If $\rho(r) = 0$ in some
connected region:

$$\nabla^2 U = 0$$

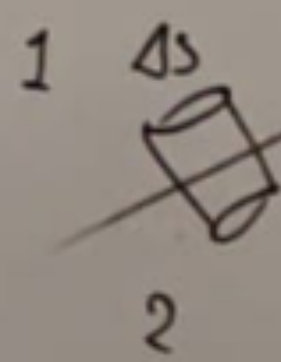


Unit 2: Electric fields in dielectric Media.



$$\vec{P}(r) = \lim_{dV \rightarrow 0} \frac{\Delta \vec{P}}{\Delta V}$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$



Boundary condition

$$\begin{aligned} D_{2n} - D_{1n} &= \sigma_f \\ E_{1t} &= E_{2t} \end{aligned}$$

Poisson and Laplace equation
for dielectric media.

$$\int_S \vec{D} \cdot \hat{n} da = Q_f = \int_V \rho(r) dv$$

$$\int_V \nabla \cdot \vec{D} dv = \int_V \rho(r) dv$$

$$\int_V (\nabla \cdot \vec{D} - \rho) dv = 0$$

Unit 2: Electric fields in dielectric Media.

Boundary conditions:

$$(a) \quad U_{out}|_{r=a} = U_{in}|_{r=a}$$

$$U(r, \theta) = \sum_{i=0}^{\infty} (A_i r^i + B_i r^{-(i+1)}) P_i(\cos \theta)$$

$$U_{in} = \sum_{i=0}^{\infty} (A_i r^i + B_i r^{-(i+1)}) P_i(\cos \theta)$$

$$U_{out} = \sum_{i=0}^{\infty} (C_i r^i + D_i r^{-(i+1)}) P_i(\cos \theta)$$

Tenemos... Dos fronteras (medios)

$$\frac{\cdot A}{\cdot B}$$

$$U_2 - U_1 = - \int_B^A \vec{E} \cdot d\vec{l}$$

$$U_2 = U_1$$

in dielectric

Boundary conditions:

(a) $U_{out}|_{r=a} = U_{in}|_{r=a}$

(b) $U_{out}|_{r=a} = -E_0 r \cos \theta$

$\vec{E} = E_0 \hat{k} = -\nabla U$

$U = -E_0 z = -E_0 r \cos \theta$

(c) $U_{in}|_{r=0} < \infty$

$P_i(\cos \theta)$

$B_i = 0$

$C_i = 0 ; i \neq 1$

$U_2 - U_1 = - \int_B^A \vec{E} \cdot d\vec{l} = 0$

$\Rightarrow U_2 = U_1$

$$d) D_{2n}|_{r=a} = D_{1n}|_{r=a}$$

$$B_i = 0$$

$$C_i = 0 ; i \neq 1$$

$$U_{in} = \sum_{i=0}^{\infty} A_i r^i P_i(\cos \theta)$$

$$U_{out} = C_1 r P_1(\cos \theta) + \sum_{i=0}^{\infty} D_i r^{-(i+1)} P_i(\cos \theta)$$

idea

$$a\hat{i} + b\hat{j} + c\hat{k} = 2\hat{i} + 3\hat{j}$$

$$\Rightarrow \begin{matrix} a=2 \\ b=3 \\ c=0 \end{matrix} \left\{ \begin{matrix} P_1(\cos \theta) \\ A P_1(\cos \theta) + B P_2(\cos \theta) \\ = 3 P_1(\cos \theta) + 2 P_2(\cos \theta) \end{matrix} \right.$$

First, using (a), we find the system of equations given by:

Tenemos posiblemente...

$$U_{in} = C_1(a) P_1 \cos \theta + D_2(a)^{-2} P_2 \cos \theta + D_3(a)^{-3} P_3 \cos \theta + D_4(a)^{-4} P_4 \cos \theta \\ + D_5(a)^{-5} P_5 \cos \theta \dots + D_n(a)^{-(n+1)} P_n \cos \theta$$

↑
Polinomio de Legendre

Tenemos

$$A_1 a = C_1 a + D_1 a^{-2} \quad \text{if } i \neq 1, \text{ then}$$

$$U_{in} = \sum_{i=0}^{\infty} A_i r^i P_i(\cos \theta)$$

$$\hookrightarrow A_i a_i = D_i a^{-(i+1)}$$

second, using (b), we obtain the following system of equations:

$$\epsilon_0 \left. \frac{\partial U_{out}}{\partial r} \right|_{r=a} = \epsilon \left. \frac{\partial U_{in}}{\partial r} \right|_{r=a}$$

↑ ↑
Frontera Frontera

Tenemos

$$\epsilon_0 \left[C_1 P_1 \cos \theta - (i+1) \sum_{i=0}^{\infty} D_i r^{-(i+2)} P_i(\cos \theta) \right] = \epsilon \left[\sum_{i=0}^{\infty} A_i r^{i-1} P_i(\cos \theta) \right]$$

Tenemos

$$\left. \varepsilon_0 \left[c, P, \cos \theta - \sum_{i=0}^{\infty} (i+1) D_i r^{-(i+2)} P_i(\cos \theta) \right] \right|_{r=a} = \left. \varepsilon \left[\sum_{i=0}^{\infty} A_i(i) r^{i-1} P_i(\cos \theta) \right] \right|_{r=a}$$

$$\Rightarrow \varepsilon_0 \left[c, P, \cos \theta - \sum_{i=0}^{\infty} (i+1) D_i a^{-(i+2)} P_i(\cos \theta) \right] = \varepsilon \left[\sum_{i=0}^{\infty} A_i(i) a^{i-1} P_i(\cos \theta) \right]$$

Desamollando la expansión...

$$\varepsilon_0 \left[c, P, \cos \theta - 1 D_0 a^{-2} P_0 \cos \theta - 2 D_1 a^{-3} P_1 \cos \theta - 3 D_2 a^{-4} P_2 \cos \theta \dots \right]$$

$$= \varepsilon \left[A_0(0) a^{-1} P_0 \cos \theta + A_1(1) a^0 P_1 \cos \theta + A_2(2) a^1 P_2 \cos \theta \dots \right]$$

$$\varepsilon_0 P_1 \cos \theta [\dots] = \varepsilon P_1 \cos \theta [\dots]$$

Tenemos

$$\varepsilon_0 [c, P, \cos \theta - 2 D_1 a^{-3} P_1 \cos \theta] = \varepsilon [A_1 P_1 \cos \theta] \Rightarrow \varepsilon_0 [c, -2 D_1 a^{-3}] = \varepsilon A_1$$

$$- \varepsilon_0 [D_0 a^{-2} P_0 \cos \theta] = 0 \Rightarrow - \varepsilon_0 D_0 a^{-2} P_0 \cos \theta = 0$$

$$- \varepsilon_0 [3 D_2 a^{-4} P_2 \cos \theta] = \varepsilon [2 A_2 a P_2 \cos \theta] \Rightarrow - \varepsilon_0 3 D_2 a^{-4} = \varepsilon 2 A_2 a$$

$$- \varepsilon_0 P_2 \cos \theta [\dots] = \varepsilon P_2 \cos \theta [\dots]$$

Tenemos ($i=1$)

$$\varepsilon A_1 = \varepsilon_0 c, -2 D_1 a^{-3}$$

$$- \varepsilon_0 (1+i) D_i a^{-(i+2)} = i \varepsilon A_i a^i$$

$i \neq 1$

$$-\epsilon_0 (1+i) D_i \bar{a}^{-(i+3)} = i \epsilon A_i a^i$$

Check Reitz - Milford

T_{area}

If $i=1, 2$ then,

$$\boxed{A_i a = C_i a + D_i \bar{a}^2} \quad \checkmark$$

If $i \neq 1, 2$ then,

$$\boxed{A_i a^i = D_i \bar{a}^{-(i+1)}} \quad \checkmark$$

Second, using (b), we obtain the following system of equations:

$$\epsilon_0 \left. \frac{\partial U_{out}}{\partial r} \right|_{r=a} = \epsilon \left. \frac{\partial U_{in}}{\partial r} \right|_{r=0} \quad \begin{cases} i=1 & \checkmark \\ i \neq 1 & \checkmark \end{cases} \quad \begin{matrix} A_i = 0; \\ D_i = 0 \end{matrix}$$